

ANSHUL RAVINDRA MANDEKAR

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Professional Summary

Computer Science engineer with experience across **Full-Stack Development**, **Data Science**, and **AI/ML**, building production systems, and data-driven applications. **1st Rank** at PBL-2 Industry Conclave 2026. Published researcher in **Generative AI and Urban Digital Twins** through an SCI-indexed publication.

Technical Skills

- **Languages:** SQL, C++, JavaScript, Python, Java, C# and dotnet (basic)
- **Web & Full-Stack:** MERN Stack (MongoDB, Express.js, React.js, Node.js), REST/Web APIs, JWT, WebSocket (Socket.IO), Mongoose, TypeScript, Vite, Tailwind CSS, Shadcn UI, React Router, React Hook Form, Zod
- **AI, ML & Generative AI:** LangChain, LangGraph, LLM Agents, LLM Fine-tuning, Prompt Engineering, RAG, ChromaDB, Transformer Architecture, CNN, ANN, YOLOv8/v9, WGAN-GP, TensorRT, Multi-modal AI, Explainable AI (XAI), Scikit-learn, NumPy, Pandas, Feature Engineering, Data Preprocessing, Model Evaluation, Streamlit
- **Data, Databases & Analytics:** MySQL, MongoDB, DBMS, Data Warehousing, Data Mining, Data Modeling, Data Access Patterns, Data Analysis, Data Visualization, Statistical Analysis, PowerBI
- **Cloud, DevOps & Architecture:** AWS (EC2, S3, ELB, ASG, Parameter Store, IAM), Docker, Kubernetes, Cloud-Native Development, Distributed Systems, Microservices, Event-Driven Architecture, Apache Kafka, Networking (TCP/IP, DNS)
- **Project & Delivery Management:** Agile, Scrum, Kanban, Sprint Planning, Backlog Grooming, JIRA, Trello, Confluence, Stakeholder Management, Cross-Functional Team Coordination, Timeline & Resource Planning, Risk Management, Requirement Gathering, Status Reporting
- **Engineering Practices & Core Competencies:** Git, Design Patterns, Code Review, Documentation, TDD, BDD, DSA (250+ LeetCode), OOPs, Problem-Solving, Analytical Thinking, Consulting Mindset, Client Communication, Ownership, Adaptability

Work Experience

Software Development Intern — MERN Stack & Cloud, Yellow Cube (Navi Mumbai) Jan 2025 – Jul 2025

- **API & Platform Engineering:** Designed and maintained **35+ RESTful API endpoints** for a role-based CRM (Node.js, Express, MongoDB/Mongoose) covering user management, permissions, reporting, and audit logs; benchmarked at **~120ms average response time** with **P95 latency under 340ms** in production.
- **Cloud Infrastructure:** Engineered and maintained scalable AWS EC2 infrastructure using ELB and Auto Scaling Groups (1–4 instances), supporting **2,000+ active users** with **90% service availability** and peak CPU utilization under 65%.
- **Security & Integration:** Implemented JWT-based authentication and RBAC; configured IAM roles, security groups, and environment secrets via AWS Parameter Store on a least-privilege model to integrate Twilio, Cloudinary, Nodemailer, and Google Sheets.
- **Real-Time Systems:** Engineered a live scheduling module using WebSockets (Socket.IO) handling **500+ concurrent appointments**, reducing booking conflicts by **60%** with sub-20ms event propagation latency.
- **Delivery & Data Management:** Followed Agile/Scrum sprints with sprint planning, backlog grooming, and stakeholder status reporting for on-time delivery; designed SQL/DBMS schemas supporting reporting and data mining for internal analytics.

Technical Projects

Explainable Financial Background Review System

Jan 2026 – Present

Python, LangChain, LangGraph, LLM Agents, RAG, ChromaDB, Streamlit, AWS EC2

- Architected a **multi-agent LLM system of 7 specialized financial agents** using LangGraph, deployed on AWS EC2 (t3.medium) achieving **~97% uptime**.
- Automated an ingestion pipeline extracting **50+ financial metrics** (CAGR, volatility, growth ratios) from Bloomberg PDFs of Fortune 500 companies at **~3.2s average parse-to-output latency**.

- Achieved a **1.00 correctness score**, 0.833 cross-reference coverage, and P95 response time under 2.8s across 12 concurrent analyst sessions — securing **1st Rank at the PBL-2 Industry Conclave 2026 (SIT Pune)**.

ISKCON Memorial Platform

Jul 2025 – Dec 2025

React.js, TypeScript, Vite, Node.js, Express.js, MongoDB, JWT, Cloudinary, Tailwind CSS

- Led development of a full-stack platform to digitally preserve and honor ISKCON devotees, shipping **12+ production REST APIs** with JWT auth (15-min access + 7-day refresh tokens), Google OAuth2 sign-in, and a 3-tier moderation workflow for **100+ users**.
- Built a responsive React/Vite/Tailwind + Shadcn UI frontend with a reusable component library, Context API state management, and advanced search/filtering by location, role, and devotional identity, improving discoverability.
- Achieved a **90+ Google PageSpeed score** via Cloudinary CDN, lazy-loading, and code-splitting, reliably handling **500+ image uploads**.

Microplastic Detection Pipeline — Symbiosis Centre for Waste Resource Management (SCWRM)

Aug

2025 – Dec 2025

YOLOv8/v9, GANs (WGAN-GP), Streamlit, TensorRT, Python

- Developed an object detection model classifying 5 microplastic categories, achieving **92.7% mAP@0.5** and outperforming ResNet/YOLOv8 baselines.
- Curated and augmented a dataset of **1,900+ annotated environmental images** using GANs to improve model robustness for small-particle detection (~50µm).

Intrusion Detection System (IDS)

Aug 2025 – Dec 2025

CNN, Feature Engineering, TensorRT, Python, NumPy, Pandas

- Built a multi-class Intrusion Detection System using a CNN on the **CICIDS2017 dataset (500,000+ records)**, featuring a novel 1D-to-2D transformation reshaping network flow data into 9×9 grayscale image matrices for spatial pattern learning.
- Designed and evaluated a Sequential CNN (dual Conv2D, MaxPooling, Dropout) with the Adam optimizer, achieving **95%+ accuracy** across all 12 attack classes, assessed via classification report, confusion matrix, ROC curves, and AUC scores.

Education

Symbiosis Institute of Technology, Pune

Aug 2023 – May 2027

B.Tech in Computer Science and Engineering — CGPA: 7.45

School of Scholars, Nagpur

Class XII (CBSE), Atrey Layout — 83% | Class X (CBSE), Beltarodi — 89%

Key Achievements

- **1st Rank**, PBL-2 Industry Conclave 2026 (SIT Pune) — multi-agent LLM system, 7 agents, 1.00 correctness score
- **SCI-indexed publication** on Urban Digital Twin — published researcher in Generative AI
- **Level 2 Qualifier**, Deloitte Hackspllosion 2026
- **200+ LeetCode DSA problems** solved

Certifications

- **Nutanix Certified Associate (NCA)** — Accelerator Program
- **Full Stack Generative AI**, Udemy (2026)
- **Machine Learning A-Z**, Udemy (2026)
- **Delta Full Stack**, Apna College (2025)
- **Community Engagement**, MichiganX/edX (2026)